LAB SHEET: WEB DESIGN TECHNOLOGY (LAB 3.2)

TITLE: CONSTRUCTING AN INTERACTIVE STUDENT PORTFOLIO WITH BOOTSTRAP 5

TOPIC: 3.2 Construct Interactive Web Elements with Bootstrap

STUDENT INFORMATION

Name	Registration No	Date
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1. LEARNING OUTCOMES

By the end of this session, you will be able to:

- **Implement** professional Bootstrap Modals for data entry (3.2.1).
- **Enhance** User Experience (UX) using interactive Tooltips and Popovers (3.2.2).
- **Construct** an Advanced Carousel featuring a custom progress bar and thumbnail navigation (3.2.3).

2. THEORETICAL OVERVIEW

To create a high-fidelity interface, we will combine Bootstrap 5 with Custom JavaScript:

- **The Modal:** A "child" window that pops up to focus user attention.
- **Popper.js Components:** Tooltips (hover) and Popovers (click). *Note: These are not auto-enabled; we must "wake them up" using JavaScript!*
- **Carousel Events:** We use the `slid.bs.carousel` event to sync our progress bar and thumbnails in real-time.

3. STEP-BY-STEP IMPLEMENTATION

Step 1: Document Setup & Design Language

Create a file named `portfolio.html`. We'll define a "Primary Purple" theme to make your portfolio look modern and premium.

Action: Setup the `<head>` and define the CSS variables:
HTML

```
<style>
    :root {
        --primary-purple: #5d46b8;
        --bg-soft: #f0f4f8;
    }
    body { background-color: var(--bg-soft); font-family: 'Segoe UI',
sans-serif; }
    .navbar-custom { background-color: var(--primary-purple); }
    .card { border: none; border-radius: 12px; box-shadow: 0 4px 10px
rgba(0, 0, 0, 0.05); margin-bottom: 20px; }

    /* Advanced Carousel Styling */
    .carousel-demo img { height: 400px; object-fit: cover; border-
radius: 8px; }
    .active-thumb { border: 3px solid var(--primary-purple)
!important; opacity: 1 !important; }
    .thumbs img { width: 100px; height: 60px; object-fit: cover;
cursor: pointer; opacity: 0.6; transition: 0.3s; }

    #btnTop { display: none; position: fixed; bottom: 30px; right:
30px; z-index: 99;
              background-color: var(--primary-purple); color: white;
border: none;
              padding: 15px; border-radius: 50%; transition: 0.3s; }
</style>
```

📸 **Checkpoint 1:** Screenshot your blank page/setup and explain why we use CSS `:root` variables.

Step 2: Profile Section (Tooltips & Popovers)

Let's add your identity! We will use **Tooltips** for skill badges and a **Popover** for your bio.

Action: Insert the Navbar and Profile Card into the `<body>`:
HTML

```
<nav class="navbar navbar-expand-lg navbar-dark navbar-custom mb-4">
    <div class="container justify-content-center">
        <ul class="navbar-nav">
            <li class="nav-item"><a class="nav-link active"
href="#"><i class="fas fa-user-circle"></i> My Profile</a></li>
            <li class="nav-item"><a class="nav-link" href="#"><i
class="fas fa-laptop-code"></i> Web Tech</a></li>
        </ul>
    </div>
</nav>

<div class="container pb-5">
    <div class="row">
        <div class="col-lg-4">
            <div class="card p-3 text-center">
                <img src="https://via.placeholder.com/120"
class="rounded-3 mb-3 shadow-sm mx-auto" alt="Profile">
                <h4 class="fw-bold mb-1">Your Name</h4>
                <p class="text-muted small">ID: 10DITXXXXXX</p>
                <div class="d-flex flex-wrap justify-content-center
gap-2 mt-2">
                    <span class="badge bg-primary" data-bs-
toggle="tooltip" title="Final Year Student">Malaysia</span>
                    <span class="badge bg-secondary" data-bs-
toggle="tooltip" title="HTML/CSS/JS">Frontend Dev</span>
                </div>
            </div>
            <div class="card p-3">
                <h5><i class="fas fa-info-circle text-primary me-
2"></i>Quick Bio</h5>
                <button type="button" class="btn btn-sm btn-outline-
primary" data-bs-toggle="popover"
                        data-bs-trigger="focus" title="Academic
Background"
                        data-bs-content="Currently pursuing Diploma
in IT.">View Detailed Bio</button>
            </div>
        </div>
```

📸 **Checkpoint 2:** Show your Profile Section. Hover your mouse over the "Malaysia" badge to show the Tooltip.

Step 3: The Advanced Project Carousel

Time to show off your work! This carousel includes a **Progress Bar** and **Thumbnails**.

Action: Add this inside the `col-lg-8` div:
HTML

```
            <div class="col-lg-8">
                <div class="card shadow-sm border-0">
                    <div class="card-body p-4">
                        <header class="mb-4 text-center">
                            <h2 class="h4 fw-bold text-
primary">Project Showcase</h2>
                            <p class="text-muted small">Hover to
pause, click thumbnails to navigate.</p>
                        </header>

                        <div id="demoCarousel" class="carousel slide
carousel-demo" data-bs-ride="carousel">
                            <div class="carousel-inner shadow-sm
rounded">
                                <div class="carousel-item active">
                                    <img
src="https://images.unsplash.com/photo-1551434678-
e076c223a692?auto=format&fit=crop&w=1200&h=600"
                                        class="d-block w-100"
alt="Database">
                                    <div class="carousel-caption d-
none d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>User Interface
Design</h5>
                                        <p>Crafting modern layouts
with Bootstrap 5 Grid and Utility classes.</p>
                                    </div>
                                </div>
```

```
<div class="carousel-item">
                                    <img
src="https://images.unsplash.com/photo-1498050108023-
c5249f4df085?auto=format&fit=crop&w=1200&h=600"
                                        class="d-block w-100" alt="JS
Interactivity">
                                    <div class="carousel-caption d-none
d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>Database Integration</h5>
                                        <p>Dynamic data management with
PHP & MySQL structural logic.</p>
                                    </div>
                                </div>
                                <div class="carousel-item">
                                    <img
src="https://images.unsplash.com/photo-1507721999472-
8ed4421c4af2?q=80&w=1200&h=600&auto=format&fit=crop"
                                        class="d-block w-100" alt="JS
Interactivity">
                                    <div class="carousel-caption d-none
d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>JavaScript Interactivity</h5>
```

```
                                        <p>Mastering DOM manipulation and
Event Listeners for seamless UX.</p>
                                    </div>
                                </div>
                            </div>
```

```
                            <button class="carousel-control-prev"
type="button" data-bs-target="#demoCarousel"
                                data-bs-slide="prev">
                                <span class="carousel-control-prev-
icon"></span>
                            </button>
                            <button class="carousel-control-next"
type="button" data-bs-target="#demoCarousel"
                                data-bs-slide="next">
                                <span class="carousel-control-next-
icon"></span>
                            </button>
                        </div>

                        <div class="progress mt-3" style="height:6px;">
                            <div id="carouselProgress"
                                class="progress-bar progress-bar-striped
progress-bar-animated bg-info"
                                style="width:0%"></div>
                        </div>

                        <div class="thumbs d-flex justify-content-center
gap-2 mt-4">
                            <img src="https://images.unsplash.com/photo-
1551434678-e076c223a692?auto=format&fit=crop&w=1200&h=600" class="img-
thumbnail active-thumb"
                                data-slide-to="0">
                            <img src="https://images.unsplash.com/photo-
1498050108023-c5249f4df085?auto=format&fit=crop&w=1200&h=600" class="img-
thumbnail" data-slide-to="1">
                            <img src="https://images.unsplash.com/photo-
1507721999472-8ed4421c4af2?q=80&w=1200&h=600&auto=format&fit=crop"
class="img-thumbnail" data-slide-to="2">
                        </div>
                    </div>
                </div>
```

```
<div class="card p-4 border-start border-info border-5 shadow-sm">
                    <div class="d-flex justify-content-between align-
items-center">
                        <div>
                            <h5 class="fw-bold mb-1">Class
Registration</h5>
                            <p class="text-muted small mb-0">Join the Web
Design Technology session for Semester 2.</p>
                        </div>
                        <button type="button" class="btn btn-info text-
white fw-bold px-4" data-bs-toggle="modal"
                            data-bs-target="#registrationModal">
                            <i class="fas fa-user-plus me-2"></i>Register
                        </button>
                    </div>
                </div>
            </div>
```

```
        </div>
    </div>
```

Step 4: Registration Modal & Footer

We will add a "Call to Action" button that opens a beautiful modal.

Action: Add the Modal at the very bottom of your `<body>`:
HTML

```
<div class="modal fade" id="registrationModal" tabindex="-1">
    <div class="modal-dialog modal-dialog-centered">
        <div class="modal-content border-0">
            <div class="modal-header bg-info text-white">
                <h5 class="modal-title">Workshop Enrollment</h5>
                <button type="button" class="btn-close btn-close-
white" data-bs-dismiss="modal"></button>
            </div>
            <form id="regForm">
                <div class="modal-body p-4">
                    <label class="form-label fw-bold">Full
Name</label>
                    <input type="text" class="form-control mb-3"
id="formName" required>
                    <label class="form-label fw-bold">Student
ID</label>
                    <input type="text" class="form-control"
placeholder="DITXXXX" required>
                </div>
                <div class="modal-footer">
                    <button type="submit" class="btn btn-info text-
white px-4">Submit Application</button>
                </div>
            </form>
        </div>
    </div>
</div>

    <button onclick="topFunction()" id="btnTop" title="Go to
top"><i class="fas fa-arrow-up"></i></button>

    <footer class="navbar-custom py-4 mt-5">
        <div class="container text-center text-white">
                    <p class="mb-1">© Copyright 2026 - JTMK
Polytechnic</p>
            <p class="mb-0" style="font-size: 0.85rem; opacity:
0.8;">Prepared by: Shu Haila Mohd Yusof (Reg. ID.:
10DIT26F1705)</p>
        </div>
    </footer>
```

Step 5: Bringing It to Life (JavaScript) This is where the magic happens! We will initialize all components.

Action: Add the script before the `</body>` tag:
JavaScript

```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>

    <script>
        (function () {
            // Initialize Tooltips & Popovers (3.2.2)
            const tooltipTriggerList = document.querySelectorAll('[data-bs-toggle="tooltip"]');
            tooltipTriggerList.forEach(el => new bootstrap.Tooltip(el));

            const popoverTriggerList = document.querySelectorAll('[data-bs-toggle="popover"]');
            popoverTriggerList.forEach(el => new bootstrap.Popover(el));

            // Advanced Carousel Logic (3.2.3)
            const carouselEl = document.querySelector('#demoCarousel');
            const intervalMs = 4000;
            const bar = document.getElementById('carouselProgress');
            const thumbs = document.querySelectorAll('.thumbs img');
             const carousel = new bootstrap.Carousel(carouselEl, { interval: intervalMs, pause:
'hover' });

            let progressTimer;
            function startProgress() {
                clearInterval(progressTimer);
                bar.style.width = '0%';
                const start = Date.now();
                progressTimer = setInterval(() => {
                    const elapsed = Date.now() - start;
                    const pct = Math.min(100, (elapsed / intervalMs) * 100);
                    bar.style.width = pct + '%';
                    if (pct >= 100) clearInterval(progressTimer);
                }, 50);
            }

            // Sync Thumbnails and Progress Bar
            carouselEl.addEventListener('slid.bs.carousel', (e) => {
                thumbs.forEach(t => t.classList.remove('active-thumb'));
                thumbs[e.to].classList.add('active-thumb');
                startProgress();
            });

            // Thumbnail Navigation
            thumbs.forEach((thumb, idx) => {
                thumb.addEventListener('click', () => carousel.to(idx));
            });

            carouselEl.addEventListener('slide.bs.carousel', () => clearInterval(progressTimer));

            startProgress(); // Start progress for the first slide

            // Back to Top logic
            let mybutton = document.getElementById("btnTop");
            window.onscroll = () => { mybutton.style.display = (window.scrollY > 100) ? "block" :
"none"; };

            // Registration Feedback
```

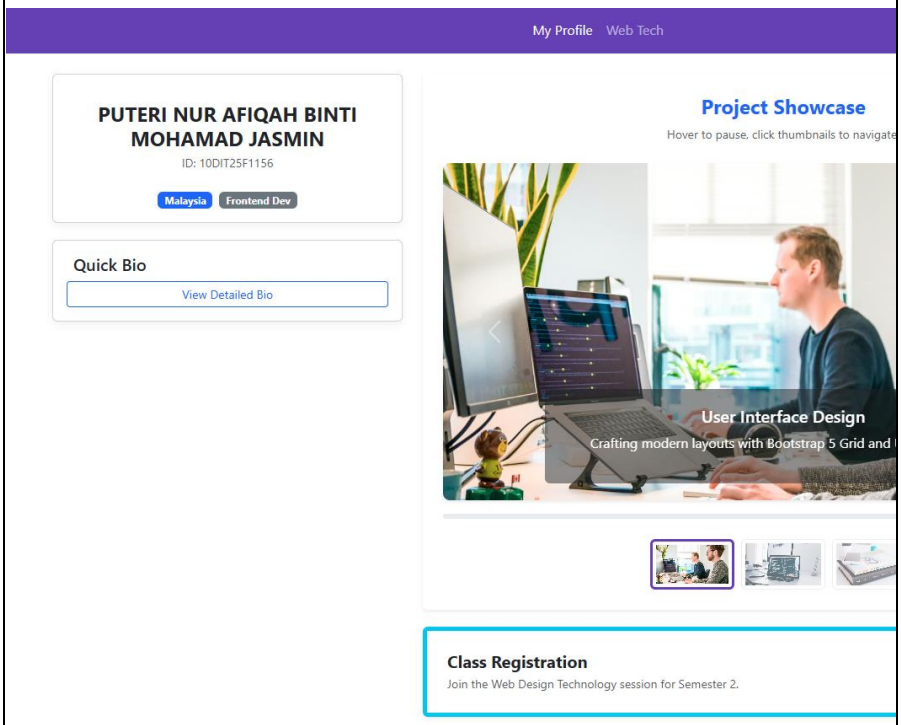
```
            document.getElementById('regForm').addEventListener('submit', function (e) {
                e.preventDefault();
                 alert("Success! Registration for " + document.getElementById('formName').value +
" is complete.");
                bootstrap.Modal.getInstance(document.getElementById('registrationModal')).hide();
            });
        })();

        function topFunction() {
            window.scrollTo({ top: 0, behavior: 'smooth' });
        }
    </script>
```

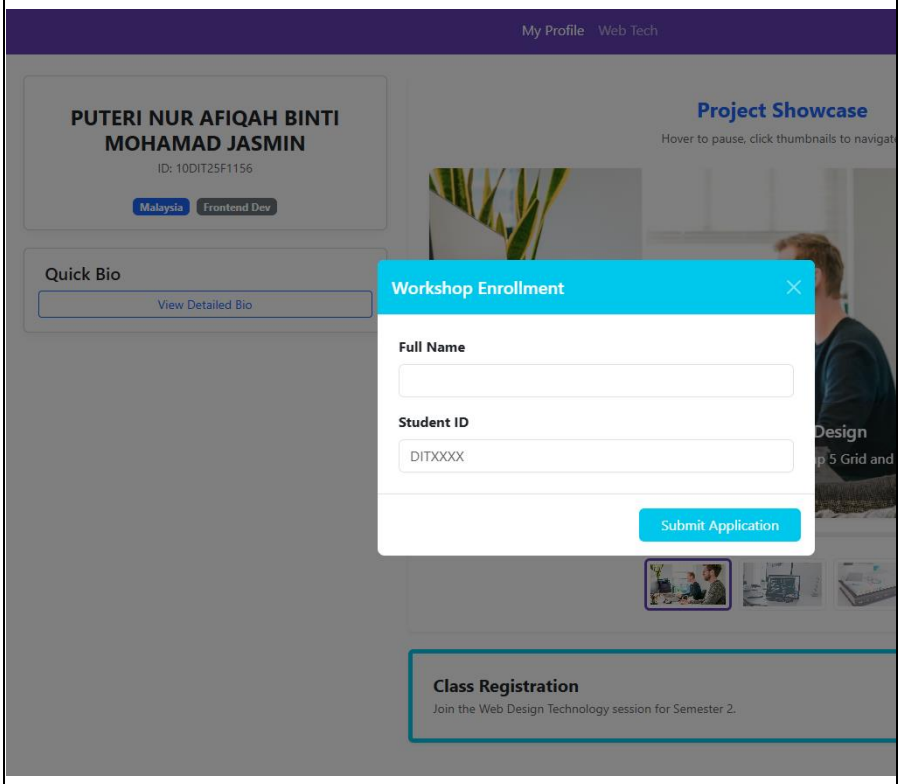
4. STEP-BY-STEP VALIDATION:

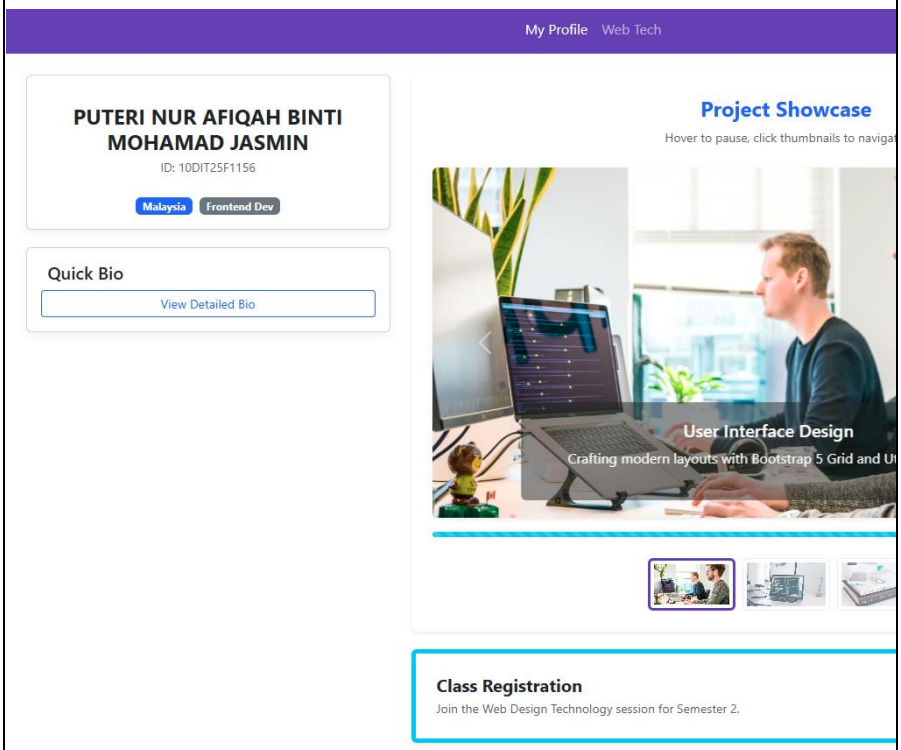
Please provide a **screenshot** for every completed step. For each image, write a short **observation** explaining the logic behind the code you just implemented. This helps solidify your mastery of Bootstrap!

Progress Step	**Screenshot of Output**	**Your Observation & Understanding**
Step 1: Setup		Setup, this part is about preparing the basic style and layout for the website. It uses CSS variables to set colors and fonts so everything looks consistent. This step also organizes how the page will be arranged, making sure the design is neat and easy to build on

Progress Step	Screenshot of Output	Your Observation & Understanding
		later. It's important because it creates a strong base for the whole website.
Step 2: Profile		the page displays the user's profile, showing their full name, ID, and their role as a frontend developer. There are navigation links at the top for easy access to different sections, and a button that allows users to view a more detailed biography. The design is clean and organized, making it easy to understand the user's basic information.

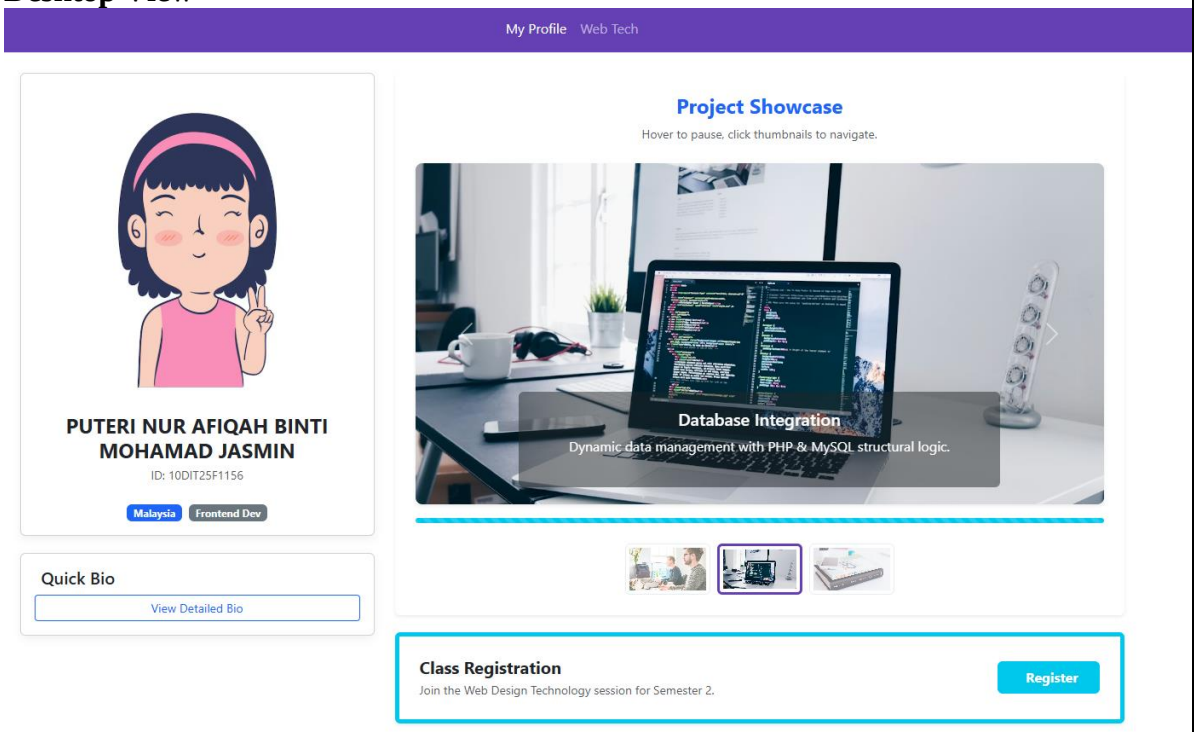
Progress Step	Screenshot of Output	Your Observation & Understanding
Step 3: The Advanced Project Carousel		the project showcase is presented as a carousel with a large main image and smaller thumbnails underneath. Users can click on the thumbnails to view different project images, which have overlay text giving brief descriptions. This interactive design helps display multiple projects neatly within a limited space.

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 4: Registration Modal & Footer		Step 4 introduces a registration modal, which pops up over the main page to allow users to enroll in a workshop. The modal contains a simple form requesting the user's full name and student ID, along with a clearly visible submit button. The background page is dimmed, focusing the user's attention on the registration form.

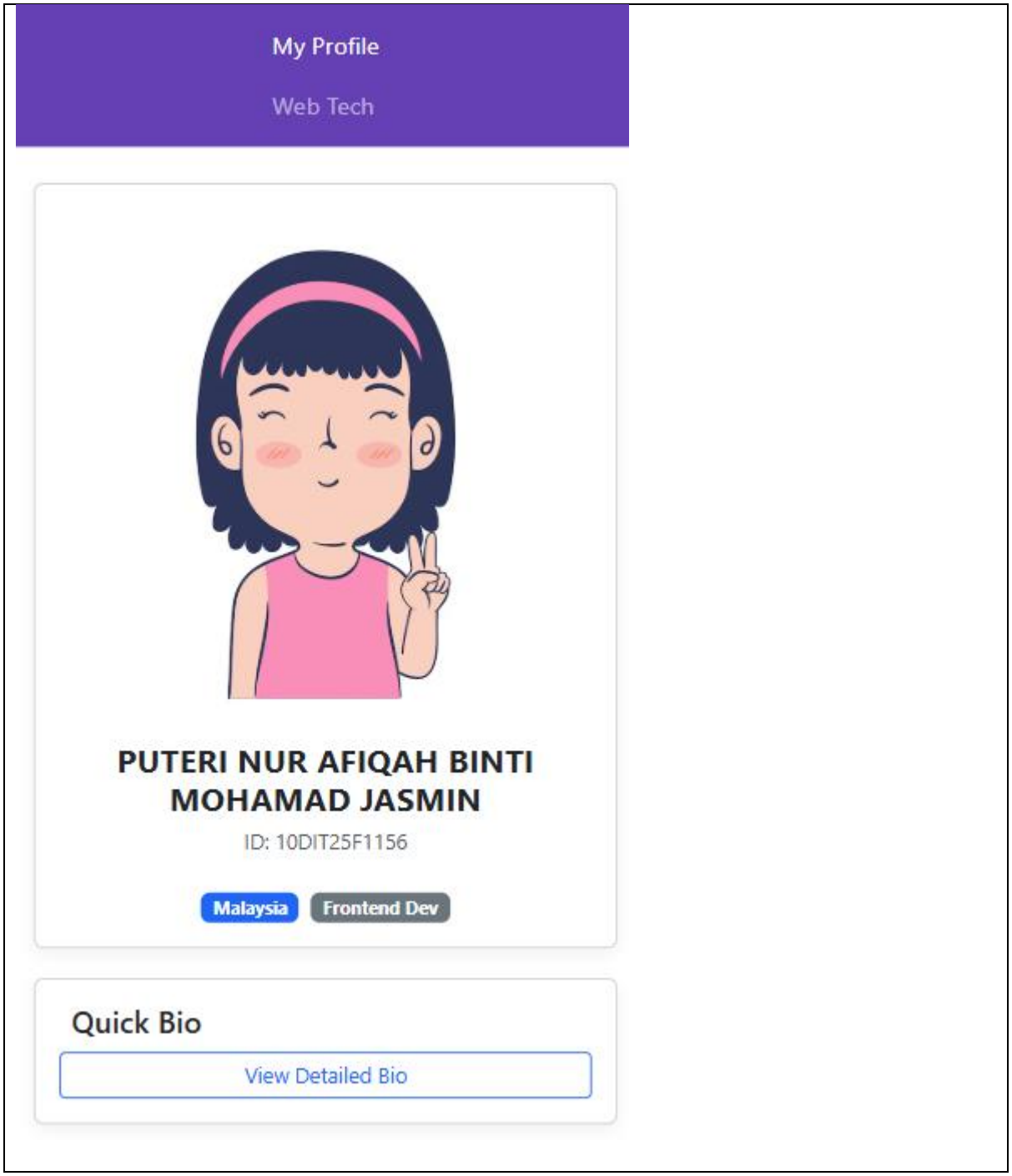
Progress Step	Screenshot of Output	Your Observation & Understanding
Step 5: Bringing It to Life (JavaScript)		JavaScript brings the carousel and modal to life by enabling smooth transitions between images and interactive buttons. The carousel automatically cycles through project images, highlighting the active thumbnail. Overall, the page looks professional and is easy to navigate, providing a good user experience.

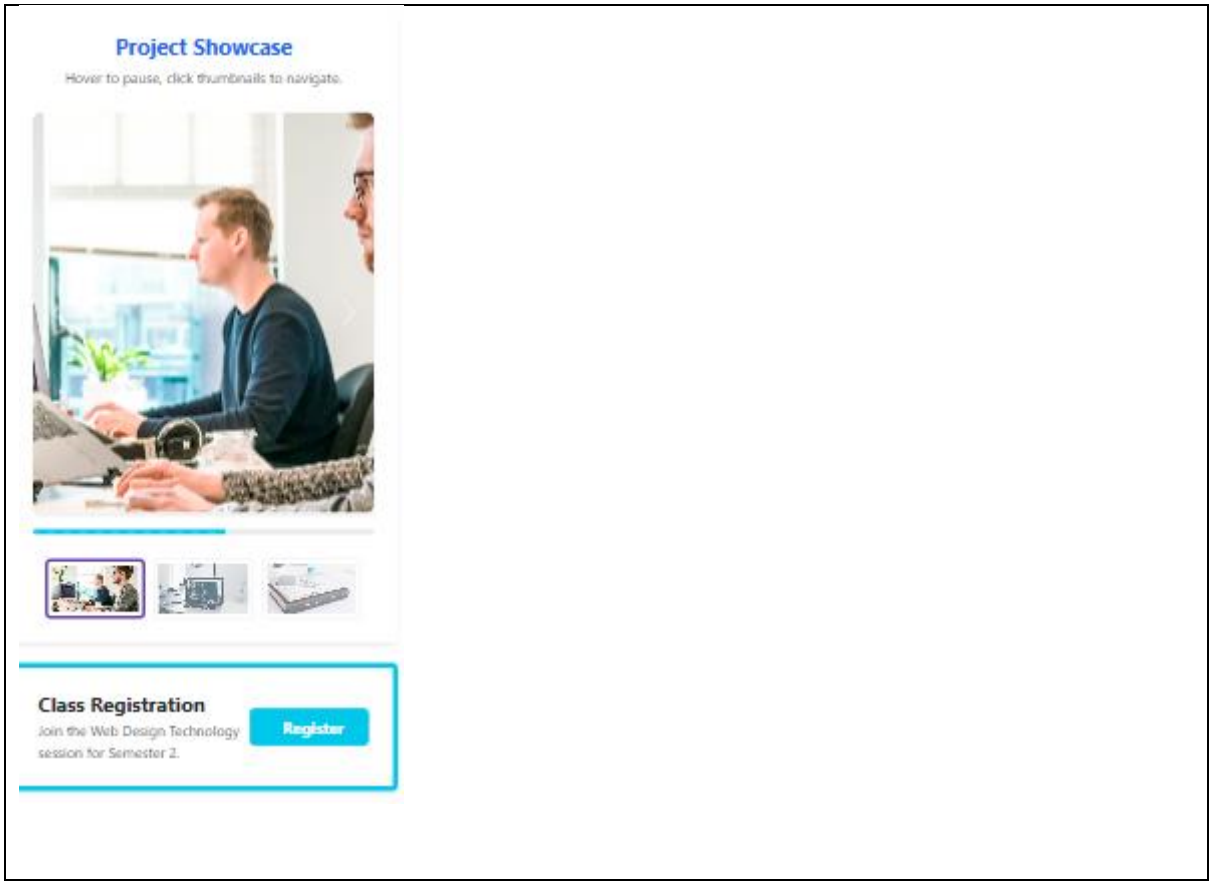
5. SCREENSHOT OUTPUT (COMPLETE):

Desktop View



Mobile View


My Profile
Web Tech
PUTERI NUR AFIQAH BINTI MOHAMAD JASMIN
ID: 10DIT25F1156
Malaysia
Frontend Dev
Quick Bio
View Detailed Bio



6. LAB SUBMISSION TASK

Ensure you have completed all steps successfully!

1. **Compress** your folder: `RegNo_Date.zip`.
2. **Report:** Create a PDF containing all screenshots and explanations: `Report_RegNo_Date.pdf`.

Happy Coding! Success is in your hands! ✨